

Twelfth Annual In-House Convention

36

24 & 25 April 1986

Room 4258/63

Department of Psychology

Dalhousie University

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Schedule

THURSDAY, APRIL 24

1:15 - 3:15

Session I. Chair: R. Klein

1. C. Baker
2. B. MacCallum and J. Barresi
3. P. McLeod
4. C. H. Jones
5. K. Bloom, A. Russell, S. Davis
6. E. Lo, K. Bloom, S. Davis
7. J. Enns

COFFEE BREAK

3:30 - 5:15

Session II. Chair: C. Reed-Elder

8. S. Kurumiya
9. A. Droungas, V. M. LoLordo
10. D. Treit
11. A. R. Delamater, D. Treit
12. R. Croll
13. J. Connolly

BEER IN LOUNGE

FRIDAY, APRIL 25

9:00 - 10:15

Session III. Hebb Symposium
Chair: M. Spetch

14. J. Fentress
15. S. Nakajima
16. D. Mitchell
17. R. Klein

COFFEE BREAK

ABSTRACTS

1. C. Baker

Lost in Space and Lost in Time

The relationship between spatial and temporal factors underlying velocity selectivity in cat striate cortex neurons was studied by obtaining responses to 2-flash stimuli for all combinations of a series of spatial and temporal separations between the two flashes. Analysis of the nonlinear components underlying the direction selectivity revealed interactions (space-time non-separability) which were generally small in magnitude, and uncorrelated in either magnitude or sign with the velocity preference measured in the same neurons. These results indicate that direction selectivity in striate cortical neurons depends separately on spatial and temporal parameters, rather than on the velocity implied by their ratio.

2. B. MacCallum and J. Barresi

Meaning in Motion: A reanalysis of the unit of perception of ongoing behavior

Three experiments were conducted in order to investigate the unitization of ongoing behavior. In the first experiment subjects were instructed to press a button while watching a film (1) "when a meaningful act ends and a different one begins", or (2) "when a meaningful act occurs". No difference was found between these instructional sets. On the whole, the results of this and two subsequent experiments, as well as, previous research suggest that people identify the meaning of events by observing rapid changes in motion that can be meaningfully interpreted rather than by identifying the beginnings and ends of hierarchically organized behavioural units.

3. P. McLeod

Early Developmental Changes in Timber Wolf Social Behaviour

The social organization of wolf packs is maintained by a complex variety of ritualized postures and actions. Here I document the early development of these postural displays plus the social interactions within which they occur. Daily video recordings of social interactions of two hand-reared wolf pups (*Canis lupus*) from eight days of age were used to document the ontogenetic emergence of the postural components of adult social interactions. I transcribed interaction sequences from video records to examine temporal patterns of pup-pup interactions. I found that postural components emerge in a consistent order, with a progressive narrowing of their contextual associations. Interactions between pups initially involved maintaining contact, but by three weeks of age, biting and pushing were common. By the fifth week, biting elicited yelps. At this time pups were also likely to raise their tails vertically before biting. At seven weeks of age, contacts became truncated. These data indicate the operation and development of processes that may be relevant to a number of mammalian social interactions.

is theoretically "strong"). Your laboratory microcomputer can help you think more clearly about these issues.

8. S. Kurumiya

Subtype of dopamine receptor crucially involved in brain stimulation reward.

The possible participation of dopaminergic neurons in intracranial self-stimulation (ICSS) has been well documented. There are two subtypes of dopamine receptors, D1 and D2. The main difference is that D1 receptors are linked to cyclic-AMP and D2 receptors are not. It is also known that neurons in the nucleus accumbens are critical in producing the rewarding effect of brain stimulation. Microinjection of SCH23390, a selective D1 receptor blocker, into the nucleus accumbens was found to suppress ICSS of the ventral tegmental area in the rat.

9. A. Droungas, V. M. LoLordo

CSc Explicitly Unpaired with the US Become Conditioned Inhibitors when CS and US Trials Occur in Single Alternation and also when they Occur in Unpredictable Order.

Two groups of rats were trained with Rescorla's (1969) "Explicitly Unpaired Procedure" (EUP) whereby US (0.5s foot-shock) and CS (60s white noise) occurred alone, separated by long intertrial intervals (III=7 min). For one group the US and CS trials occurred in single alternation (EUPs), while for the second the sequence of US and CS trials was randomized (EUPr). Strong conditioned inhibition accrued to both the EUPs- and EUPr-trained CSs as indexed by both summation and retardation tests and compared to a control group. This result, although in accord with Rescorla's (1969) theoretical formulations on conditioned inhibition resulting from negative CS-US contingency, is contrary to recent demonstrations by Fowler (1985) that have followed from Wagner's SOP theory.

10. D. Treit

Pharmacological evidence for a continuum between anxiety and convulsive states

A variety of clinical observations have suggested that some convulsive, epileptic states are correlated with "fear" or "anxiety". One specific hypothesis that can be derived from these observations is that "anxiety" might represent a mild, or "sub-clinical" epileptic state; i.e., anxiety and convulsions might lie along the same mechanistic continuum. Consistent with this speculation, many potent anti-anxiety drugs also are potent anti-convulsants, and some potent convulsant drugs appear to produce "anxiety". These pharmacological relationships may not be coincidental, and there is evidence that all of these drugs may act upon a similar neurological substrate. My research on the interactive effects of convulsants and anxiolytics within an animal model further supports the notion of a link between anxiety and convulsant states.

capture the point, which will be discussed briefly in light of some current perspectives.

15. S. Nakajima

Cell assembly: Hebb's neuronal model of learning

One of the basic assumptions in the cell assembly hypothesis is that synaptic connections are changeable. Studies in our Department by Cynader, Yoon, and Goddard have all supported this assumption. Experiments on invertebrates provides further insight into the nature of synaptic change: cyclic AMP plays an important role in producing lasting modifications of synaptic functions. Recent findings that the type of dopamine receptors which is critical in producing reinforcement effect is the D1 type -- the type that increases the production of cyclic AMP suggest a future direction of the development of the model.

16. D. Mitchell

Perceptual Development

17. R. Klein

Hebb's contribution to contemporary cognitive psychology

Through his effective attack on radical behaviorism, his unflagging emphasis on internal mental processes (thought), his cogent demonstration of weaknesses in some key assumptions of the Gestalt and S-R frameworks, and his own cell-assembly proposal, Hebb may be said to have plowed the soil from which the renaissance in cognitive psychology would grow & flourish.

18. A. Gilani and J. Enns

Object Superiority Effect as a Function of Diss & Dimensionality

The identification of a briefly presented line segment is sometimes facilitated by embedding the target line in a context of other noninformative lines (the "object superiority" and "object line" effects). This study compared two alternative hypotheses for these effects. One hypothesis claims that line figures with interpretations as three-dimensional objects are processed more rapidly and accurately by the visual system than two-dimensional figures (Weisstein and Harris, 1974). The second hypothesis claims only that some lines-in-context are more discriminable in a brief flash than others because of emergent features created by the interaction of target and context lines (Enns and Prinzmetal, 1984).

19. G. Eskes, T. McCormick and S. Bryson

Autistic children's comprehension of single words.

There is considerable evidence that children with early childhood autism suffer a general language impairment. This impairment is manifested as a difficulty in understanding the meanings of words and in organizing and

also illustrates a general principle of human and animal neural behavior which is "Show me something I don't already know".

23. L. Donovan & S. Shaw

Efferent control of visual sensitivity?

Some sensory systems receive efferent innervation, which controls their responsiveness to appropriate incoming sensory stimuli. In insects, efferent neurones run out to the eye from the brain, and we speculate that these could control the rate of visual dark adaptation, following exposure to light. The largest efferent contains the biogenic amine serotonin (5-HT, a known neurotransmitter). We have re-examined dark adaptational recovery in the photoreceptors and postsynaptic neurones, and uncovered evidence of at least 4 recovery processes (2 slow). Only one of these (fast) is described in the literature on photoreceptors. Preliminary results with small pressure injections of 5-HT into the eye show that 5-HT does depress visual sensitivity.

24. D. B. Henken

Changes in cell proliferation in the goldfish retina following optic nerve crush.

In goldfish, new cells are continuously added to the retina throughout life. Optic nerve axotomy modulates this ongoing histogenesis. Labelling densities in the ganglion cell/optic fiber layer and the photoreceptor layer are affected following unilateral optic nerve crush, when fish are allowed to survive for 1 day following injection of tritiated thymidine.

25. L. McCook

The Electroretinogram of Insect Compound Eye: The origins of the ERG, and its use in examining the determination of contrast coding in the visual system.

The cellular origins of the electroretinogram (ERG) of the blowfly compound eye were investigated for both waveform and stimulus-response characteristics. The results suggest that the ERG may be due to a summation of the responses of the photoreceptors and the lamina large monopolar cells. The ERG was also used to study the determination of the monopolar cell response to visual contrast, using flies reared under abnormal contrast conditions. The results suggest that there may be some visual plasticity, or environmental regulation of contrast coding in the visual system, which is contrary to expectations for a short lived animal.

26. F. A. Baker, M. G. Yoon, and L. I. Benowitz

The Optic Tectum Influences Production and Transport of Proteins in Retinal Ganglion Cells of Adult Goldfish.

Ganglion cells in the goldfish retina produce proteins which are transported through optic fibers to their target tissue, the optic tectum. During regeneration of optic fibers the presence of the optic tectum enhances the production of rapidly-transported proteins with molecular

lesion results, in short, indicate that the hypothalamus can take a licking, but the FED keeps on ticking. Our current efforts to localize the FED by assessing temporal variations of local cerebral 2-deoxyglucose uptake will also be discussed.

30. R. T. Ross and V. M. LoLordo*

Behavior is in the eye of the Beholder: Differences in Response Definition can Engender Diverse Theoretical Interpretations

Two procedures for observing rearing behavior during a localized visual CS paired with food were compared within the same subjects. Using the method and response definition described by Holland, rearing was acquired during conditioning trials and declined with introduction of extinction. Using the method and response definition described by Kaye and Pearce, rearing declined from baseline levels during conditioning but was temporarily restored with extinction. These within-subject outcomes indicate that the response definitions used by Holland and by Kaye and Pearce, which differ along a continuum of "directedness", assess different components of behavior occurring in response to a visual CS reflecting associative strength or associability, respectively. Thus, differences in response definition, if not recognized, can give rise to theoretical conceptions of the processes that mediate overt behavior which may appear contradictory.

* With thanks to Cheryl Reed Elder and Anastasia Droungas for serving as additional observers.

31. G. Prusky

Muscarinic and nicotinic acetylcholine binding sites in the developing cat visual cortex.

We have demonstrated that at distinctly different periods during development, both muscarinic and nicotinic acetylcholine receptors are specifically localised to layer IV of primary visual cortex. The muscarinic receptors exemplify this pattern only during early postnatal life but the nicotinic receptors develop this characteristic during the physiologically defined critical period and express it into adulthood. The different anatomical distributions of muscarinic and nicotinic receptors in the visual cortex together with other physiological evidence suggests distinctive roles for each of these receptor systems in the control of plasticity.

32. D. E. Mitchell, K. M. Murphy, H. Dzioba and J. A. Horne

The requirements for optimum behavioural recovery from the effects of early monocular deprivation in kittens.

Last year we described a procedure that permitted an amazing degree of recovery from the very severe behavioural effects of early monocular deprivation. The method combined daily periods of reverse occlusion (patching) where the kitten is forced to employ its formerly deprived eye, with periods of binocular visual exposure. In kittens monocularly deprived to 6 weeks of age, the vision of the formerly deprived eye recovered to normal levels if the formerly nondeprived eye was patched for

Faps, Maps and Saps: Contextual Rules of Integrated Action

Ethologists have documented, in behavioural terms, the existence of numerous "fixed action patterns" (FAPS) for diverse animal species, have more recently liberalized notions of extreme motor stereotypy in terms of "modal action patterns" (MAPS), and have argued for the necessary existence of independently ordered "specific action potentials" (SAPS). However, a limit to current models is an appropriate documentation of variations in the form and central control of movement sequences as a function of the broader contexts within which these movements occur. I thus report upon contextual modulations of integrated movement in rats, with and without specific sensory & central lesions, based upon collaborative work with Kent Berridge. These investigations suggest new ways to conceptualize the neural control of integrated behavioural patterns in mammals.

ADDENDUM

37. OLFACTORY RECOGNITION OF CONGENIC STRAINS OF RATS.
Richard E. Brown¹, Prim B. Singh², and Bruce J. Roser².
(¹Subdepartment of Animal Behaviour, Madingley and ²Institute of Animal Physiology, Babraham, Cambridge, England).

Recognition of different major histocompatibility (MHC) types of mice is based on urinary chemosignals from congenic strains which differ only in the MHC (Yamazaki, et al, 1979, J. Exp. Med., 150, 755-760). Our experiment provides the first evidence that congenic strains of rats which differ only in the MHC also produce different urinary chemosignals. Urine from adult male PVG and PVG-R1 rats was used in a habituation - dishabituation test (Sundberg, et al, 1982, Behav. Neur. Biol., 34, 113-119) with male PVG-RT1^u rats as subjects.

Urine from PVG males was easily distinguished from that of PVG-R1 males ($p < .001$). Individual PVG males were not distinguished by their urinary odours, but individual PVG-R1 males did produce discriminably different odours ($p < .05$). These results demonstrate that PVG and PVG-R1 males, which differ only in the A region of the MHC, produce different urinary chemosignals and that individual males of the PVG-R1 strain also produce distinguishable "individual" odours. These may be related to individual differences in the quantity of A region classical class I antigens in the urine of PVG-R1 rats (Singh and Roser, unpublished observations). We are now using immunological assays to determine how the quality and quantity of class I antigens or their fragments in the urine of these rats enables recognition of their substrain and of individuals within a highly inbred substrain.